







# Examples of f-Divergences

|                       | Entropy (generator) Function                                           | Divergence                                                                                                                                 |
|-----------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Kullback-Leibler (KL) | $f(s) = s \log s - s + 1$                                              | $\int \log\left(\frac{d\alpha}{d\beta}\right) d\alpha(x)$                                                                                  |
| Total Variation (TV)  | $f(s) =  s - 1 $                                                       | $\int  d\alpha - d\beta $                                                                                                                  |
| Jensen-Shannon        | $f(s) = \frac{1}{2}\left(s \log s - (s + 1)\log\frac{s + 1}{2}\right)$ | $\frac{1}{2} \int \log\left(\frac{d\alpha}{d\beta}\right) d\alpha(x) + \frac{1}{2} \int \log\left(\frac{d\beta}{d\alpha}\right) d\beta(x)$ |
| Hellinger             | $f(s) = (\sqrt{s} - 1)^2$                                              | $\int  \sqrt{d\alpha} - \sqrt{d\beta} ^2$                                                                                                  |
| Pearson $\chi^2$      | $f(s) = (s - 1)^2$                                                     | $\int \frac{ d\alpha - d\beta ^2}{d\beta}$                                                                                                 |
| Neyman $\chi^2$       | $f(s) = \frac{(1 - s)^2}{s}$                                           | $\int \frac{ d\alpha - d\beta ^2}{d\alpha}$                                                                                                |

# Entropy Functions

